

Labsheet - 8

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix,
classification_report

df = pd.read_csv("/content/sample_data/WA_Fn-UseC_-Telco-Customer-Churn.csv")

print(df.head())
print(df.shape)

customerID gender SeniorCitizen Partner Dependents tenure PhoneService \
0 7590-VHVEG Female 0 Yes No 1 No
1 5575-GNVDE Male 0 No No 34 Yes
2 3668-QPYBK Male 0 No No 2 Yes
3 7795-CFOCW Male 0 No No 45 No
4 9237-HQITU Female 0 No No 2 Yes

MultipleLines InternetService OnlineSecurity ... DeviceProtection \
0 No phone service DSL No ... No
1 No DSL Yes ... Yes
2 No DSL Yes ... No
3 No phone service DSL Yes ... Yes
4 No Fiber optic No ... No

TechSupport StreamingTV StreamingMovies Contract PaperlessBilling \
0 No No No Month-to-month Yes
1 No No No One year No
2 No No No Month-to-month Yes
3 Yes No No One year No
4 No No No Month-to-month Yes

PaymentMethod MonthlyCharges TotalCharges Churn
0 Electronic check 29.85 29.85 No
1 Mailed check 56.95 1889.5 No
2 Mailed check 53.85 108.15 Yes
3 Bank transfer (automatic) 42.30 1840.75 No
4 Electronic check 70.70 151.65 Yes

[5 rows x 21 columns]
(7043, 21)

print(df.info())
print(df.isnull().sum())

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
# Column Non-Null Count Dtype
---  ---
0 customerID 7043 non-null object
1 gender 7043 non-null object
2 SeniorCitizen 7043 non-null int64
3 Partner 7043 non-null object
4 Dependents 7043 non-null object
5 tenure 7043 non-null int64
6 PhoneService 7043 non-null object
7 MultipleLines 7043 non-null object
8 InternetService 7043 non-null object
9 OnlineSecurity 7043 non-null object
10 OnlineBackup 7043 non-null object
```

11 DeviceProtection 7043 non-null object
12 TechSupport 7043 non-null object
13 StreamingTV 7043 non-null object
14 StreamingMovies 7043 non-null object
15 Contract 7043 non-null object
16 PaperlessBilling 7043 non-null object
17 PaymentMethod 7043 non-null object
18 MonthlyCharges 7043 non-null float64
19 TotalCharges 7043 non-null object
20 Churn 7043 non-null object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
None
customerID 0
gender 0
SeniorCitizen 0

Partner 0
Dependents 0
tenure 0
PhoneService 0
MultipleLines 0
InternetService 0
OnlineSecurity 0
OnlineBackup 0
DeviceProtection 0
TechSupport 0
StreamingTV 0
StreamingMovies 0
Contract 0
PaperlessBilling 0
PaymentMethod 0
MonthlyCharges 0
TotalCharges 0
Churn 0
dtype: int64

```
# Remove customer ID if present
if "customerID" in df.columns:
    df = df.drop("customerID", axis=1)
```

```
print(df.head())
```

```
gender SeniorCitizen Partner Dependents tenure PhoneService \
0 Female 0 Yes No 1 No
1 Male 0 No No 34 Yes
2 Male 0 No No 2 Yes
3 Male 0 No No 45 No
4 Female 0 No No 2 Yes
```

```
MultipleLines InternetService OnlineSecurity OnlineBackup \
0 No phone service DSL No Yes
1 No DSL Yes No
2 No DSL Yes Yes
3 No phone service DSL Yes No
4 No Fiber optic No No
```

```
DeviceProtection TechSupport StreamingTV StreamingMovies Contract \
0 No No No No Month-to-month
1 Yes No No No One year
2 No No No No Month-to-month
3 Yes Yes No No One year
4 No No No No Month-to-month
```

```
PaperlessBilling PaymentMethod MonthlyCharges TotalCharges \
0 Yes Electronic check 29.85 29.85
1 No Mailed check 56.95 1889.5
2 Yes Mailed check 53.85 108.15
3 No Bank transfer (automatic) 42.30 1840.75
```

4 Yes Electronic check 70.70 151.65

Churn

0 No

1 No

2 Yes

3 No

4 Yes

```
df = df.dropna()
```

```
print("Missing values:")
```

```
print(df.isnull().sum())
```

Missing values:

gender 0

SeniorCitizen 0

Partner 0

Dependents 0

tenure 0

PhoneService 0

MultipleLines 0

InternetService 0

OnlineSecurity 0

OnlineBackup 0

DeviceProtection 0

TechSupport 0

StreamingTV 0

StreamingMovies 0

Contract 0

PaperlessBilling 0

PaymentMethod 0

MonthlyCharges 0

TotalCharges 0

Churn 0

dtype: int64

```
# Convert categorical columns into numerical values
```

```
df = pd.get_dummies(df, drop_first=True)
```

```
print(df.head())
```

SeniorCitizen tenure MonthlyCharges gender_Male Partner_Yes \

0 0 1 29.85 False True

1 0 34 56.95 True False

2 0 2 53.85 True False

3 0 45 42.30 True False

4 0 2 70.70 False False

Dependents_Yes PhoneService_Yes MultipleLines_No phone service \

0 False False True

1 False True False

2 False True False

3 False False True

4 False True False

MultipleLines_Yes InternetService_Fiber optic ... TotalCharges_996.45 \

0 False False ... False

1 False False ... False

2 False False ... False

3 False False ... False

4 False True ... False

TotalCharges_996.85 TotalCharges_996.95 TotalCharges_997.65 \

0 False False False

1 False False False

2 False False False

3 False False False

4 False False False

```
TotalCharges_997.75 TotalCharges_998.1 TotalCharges_999.45 \
0 False False False
1 False False False
2 False False False
3 False False False
4 False False False
```

```
TotalCharges_999.8 TotalCharges_999.9 Churn_Yes
0 False False False
1 False False False
2 False False True
3 False False False
4 False False True
```

```
[5 rows x 6560 columns]
```

```
X = df.drop("Churn_Yes", axis=1)
y = df["Churn_Yes"]
```

```
print("Features:", X.shape)
print("Target:", y.shape)
```

```
Features: (7043, 6559)
Target: (7043,)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)
```

```
print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)
```

```
Training data: (5634, 6559)
Testing data: (1409, 6559)
```

```
model = DecisionTreeClassifier(
    criterion="gini",
    max_depth=5,
    random_state=42
)
```

```
model.fit(X_train, y_train)
```

```
print("Decision Tree model trained successfully.")
```

```
Decision Tree model trained successfully.
```

```
y_pred = model.predict(X_test)
```

<https://colab.research.google.com/drive/14ecBM9ORmfq0L8W3YjsKorChyWt9-1Gx#scrollTo=TmZqi43rHu8x&printMode=true> 3/6

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```
print("Predicted values:")
print(y_pred[:20])
```

```
Predicted values:
```

```
[ True False False True False False False False False False False
  False False False True False True False False]
```

```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
```

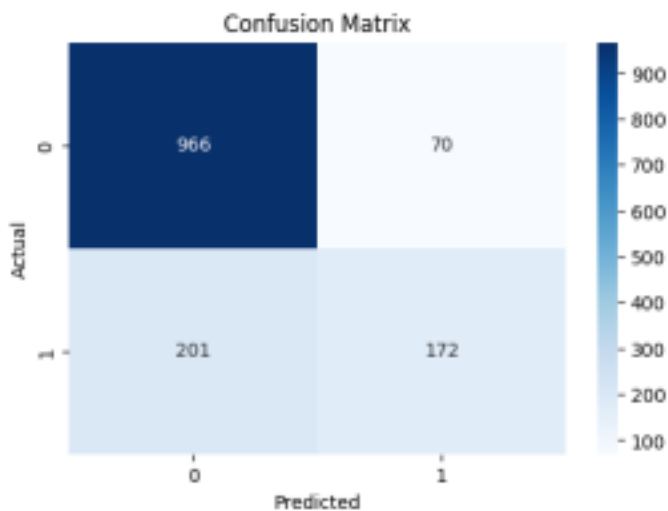
```
print("Accuracy :", accuracy)
print("Precision:", precision)
print("Recall :", recall)
print("F1-Score :", f1)
```

Accuracy : 0.8076650106458482
Precision: 0.7107438016528925
Recall : 0.46112600536193027
F1-Score : 0.5593495934959349

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



```
print(classification_report(y_test, y_pred))
```

precision recall f1-score support

False 0.83 0.93 0.88 1036

True 0.71 0.46 0.56 373

accuracy 0.81 1409

macro avg 0.77 0.70 0.72 1409

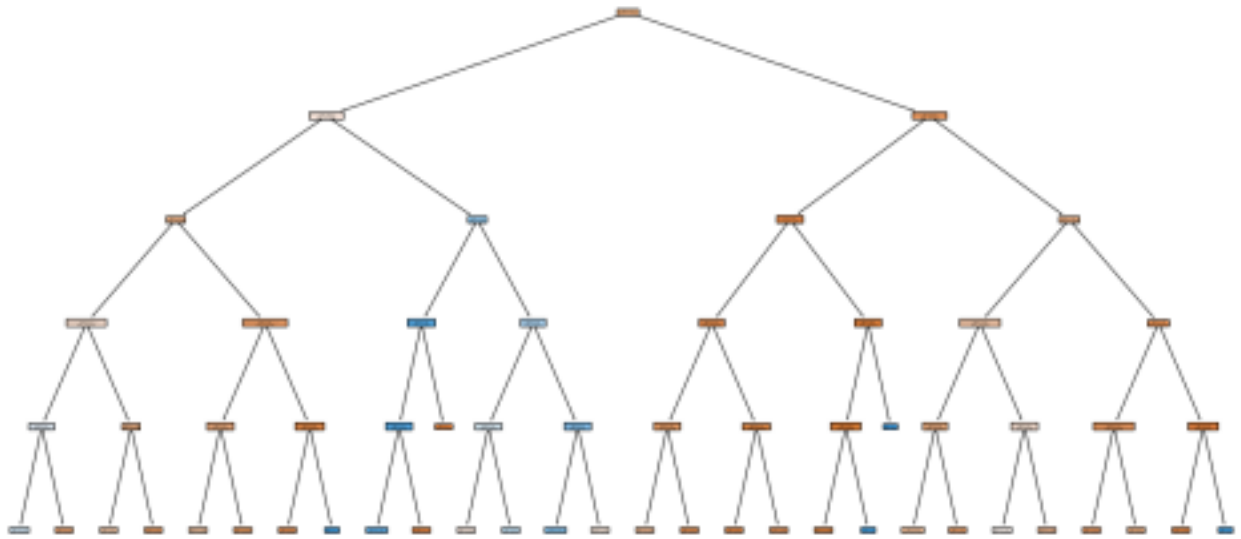
weighted avg 0.80 0.81 0.79 1409

```
plt.figure(figsize=(20, 10))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=["No Churn", "Churn"],
    filled=True,
    rounded=True
)

plt.title("Decision Tree for Customer Churn Prediction")
plt.show()
```

Decision Tree for Customer Churn Prediction



```
importance = pd.Series(
    model.feature_importances_,
    index=X.columns
).sort_values(ascending=False)

print("Important Features:")
print(importance.head(10))

plt.figure(figsize=(10, 5))
importance.head(10).plot(kind="bar")

plt.title("Top Features Influencing Customer Churn")
plt.xlabel("Features")
plt.ylabel("Importance")
plt.xticks(rotation=45)
plt.show()
```

```
Important Features:
tenure 0.482288
InternetService_Fiber optic 0.343501
PaymentMethod_Electronic check 0.027378
TechSupport_No internet service 0.022246
MonthlyCharges 0.017803
Contract_One year 0.016576
StreamingMovies_No internet service 0.015926
OnlineSecurity_Yes 0.015608
MultipleLines_Yes 0.014703
Contract_Two year 0.012857
dtype: float64
```



Theory Questions

1. What is classification, and how does it differ from regression?

Classification is a supervised learning technique used to predict **categories or classes**, such as Yes/No or Churn/Not Churn.

Regression is used to predict a **continuous numerical value**, such as salary, price, or temperature.

Example:

- Classification → Customer will churn: Yes/No
- Regression → Predict customer spending amount.

2. Explain the working principle of the Decision Tree algorithm.

A Decision Tree makes predictions by dividing the dataset into smaller groups using a series of **decision rules**.

The algorithm:

1. Selects the best feature for splitting.
2. Divides the data into branches.
3. Repeats the process for each branch.
4. Stops when a suitable stopping condition is reached.
5. Uses the final leaf node to make the prediction.

It uses criteria such as **Gini Index or Entropy** to select the best split.

3. Differentiate between Gini Index and Entropy as splitting criteria.

Gini Index	Entropy
Measures impurity in a dataset.	Measures uncertainty or disorder.
Generally faster to calculate.	Uses logarithmic calculations.
Lower Gini means a better split.	Lower Entropy means a better split.
Commonly used with Decision Trees.	Used for information-gain based splitting.
Both are used to select suitable splits in a Decision Tree.	

4. What is a Confusion Matrix? Explain its components.

A **Confusion Matrix** is a table used to evaluate the performance of a classification model.

Its four components are:

- **True Positive (TP):** Correctly predicted positive cases.
- **True Negative (TN):** Correctly predicted negative cases.
- **False Positive (FP):** Negative case incorrectly predicted as positive.
- **False Negative (FN):** Positive case incorrectly predicted as negative.

It helps understand the types of errors made by the model.

5. Define Accuracy, Precision, Recall, and F1-Score. Why are these metrics important?

- **Accuracy:** Percentage of total predictions that are correct.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

- **Precision:** Percentage of predicted positive cases that are actually positive.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

- **Recall:** Percentage of actual positive cases correctly identified.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

- **F1-Score:** Harmonic mean of Precision and Recall.

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

These metrics are important because they provide a detailed view of classification performance rather than depending only on accuracy.

6. What is overfitting in a Decision Tree? How can it be reduced?

Overfitting occurs when a Decision Tree learns the training data too closely, including noise and unnecessary details. As a result, it performs well on training data but poorly on new data.

It can be reduced by:

- Limiting the **maximum tree depth**.
- Setting a minimum number of samples for splitting.
- Pruning unnecessary branches.
- Using suitable train-test splitting and validation.

7. Why is customer churn prediction important for businesses?

Customer churn prediction helps businesses identify customers who are likely to **stop using their services or products**.

Early identification allows companies to:

- Provide targeted offers.
- Improve customer satisfaction.
- Take retention actions.
- Reduce revenue loss.

Thus, churn prediction supports better customer retention strategies.

8. What are the advantages and limitations of the Decision Tree algorithm?

Advantages:

- Easy to understand and interpret.
- Can represent decisions using simple rules.
- Useful for classification problems.
- Can identify important features.

Limitations:

- Can easily overfit the training data.
- Small changes in data can change the tree structure.
- Very large trees can become difficult to interpret.
- Performance may be lower than some advanced ensemble methods.

9. Mention three real-world applications of Decision Tree Classification other than customer churn prediction.

Three applications are:

1. **Medical diagnosis** – Predicting whether a patient has a particular disease.
2. **Credit risk analysis** – Classifying customers as low-risk or high-risk.
3. **Fraud detection** – Identifying transactions as legitimate or potentially fraudulent.

10. How can churn prediction insights help organizations improve customer retention and profitability?

Churn prediction identifies customers who have a high probability of leaving. Organizations can use these insights to provide **personalized offers, discounts, better support, and loyalty programs**.

It also helps businesses identify the factors responsible for churn and improve their services accordingly. This can reduce customer loss, increase retention, and ultimately improve **business profitability**.